Sierra Powell

762-383-5755 | spowell8986@tuskegee.edu |www.linkedin.com/in/sierra-powell-b6996526a

U.S. Citizen | Active Interim Secret Clearance

Education

Tuskegee University – Tuskegee, AL
Graduating in Fall 2027
Master of Science in Information Systems & Computer Security

Tuskegee University | Tuskegee, AL
Class of 2025
Bachelor of Science in Computer Science, GPA:3.6

Professional Experience

General Infomatics (assigned to NAVAIR Patuxent River, MD) | June 2025 – August 2025
IT & Cybersecurity Intern
• Hardened Windows, Ubuntu, and RHEL systems in air-gapped environments for Authorization to Operate (ATO).
• Applied DoD STIGs (Security Technical Implementation Guides) to all systems using STIG Viewer, improving compliance posture and reducing vulnerabilities.
• Conducted vulnerability scanning and remediation using SCAP and ACAS tools.
• Installed Windows 11 on unsupported hardware by modifying BIOS, enabling TPM/Secure Boot, and integrating drivers
• Built secure systems from the ground up: BIOS setup, disk partitioning, OS installation, and offline patching.
• Supported Risk Management Framework (RMF) documentation with clear and auditable test procedures.

CHITRA Productions (assigned to NAVAIR, Patuxent River, MD) | June 2024 – August 2024
Software Developer Intern
• Designed and developed a new feature for the acoustic software testing application, automating the extraction and saving of test configurations from the GUI into development files.
• My feature reduced the process from 30 minutes to 5 minutes per test and eliminated 90% of manual configuration errors.
• Implemented techniques such as file reading/writing, conditionals, and data manipulation to improve workflow and accuracy in configuration handling.

Awards

• Upsilon Pi Epsilon Honor Society: International Honor Society for Computing and Information disciplines.
• CyberCorps Scholarship for Service (SFS) Scholar
• MAKERS S-STEM Scholar

Projects

IoT Intrusion Detection System | Tuskegee, AL | April 2026

- Built and deployed a machine learning-based IDS using Support Vector Machine (SVM) achieving 98.88% accuracy on network intrusion detection.
- Trained model on NSL-KDD dataset classifying normal vs. anomalous traffic with 0.9895 F1 score across 7,558 test samples.
- Integrated IDS into a live IoT pipeline using NodeMCU ESP8266, Node.js backend, and Python Flask API for real-time attack detection.

ESP8266 IoT Purple Team Lab | Personal Project | April 2026

- Deployed a NodeMCU ESP8266 as a simulated IoT target, exposing HTTP and TCP services to replicate misconfigurations.
- Performed red team reconnaissance via Kali Linux OS using Nmap including port scanning, service fingerprinting, and banner grabbing, identifying version disclosure and unauthenticated data leakage.
- Hardened the device by removing verbose headers, generic banners, and eliminating unauthenticated commands, then verified all fixes by re-scanning and producing a formal security assessment report.

Skills

- Linux
- Computer Networking
- OS Deployment
- Python, C++ Programming, Java, Bash
- Version Control (Git/GitHub)
- System Auditing
- Database Management
- Security Policy & Risk Assessment